

MA1002 — Calculus for Science Part 2

Summer 2023 Examination · Paper 1 · Full Marking Scheme

Module	MA1002 — Calculus for Science Part 2
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Institution	University College Cork
Session	Summer 2023, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer 4 of the 5 questions . All questions carry equal marks (17.5 each, 70 in total). Marks for each part are shown in brackets.
Materials	Formula sheet and log tables supplied. Show all work and say where a calculator has been used.

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

Coverage note. On the day you answer 4 questions, but this scheme works through **all five** in full, so you can pick the ones you’re strongest on.

Source basis. Every answer uses MA1002 course methods: indefinite integrals, substitution, integration by parts, area, volumes of rotation, partial fractions and separable differential equations. Two parts go outside the listed topics and carry a note: Q3(b) (tangent slope) and Q5(b) (regression line, formulas supplied).

Mark splits. Part totals are the paper’s own. The step-by-step split within each part is Study-Uni’s suggested allocation.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Rounding. Answers are given exactly (e.g. $1/12$, $625\pi/3$). Q5(b) is in fractions as the question asks.

Question 1

Antiderivatives (the indefinite integral), initial conditions — 17.5 marks

1(a)

[9.5 marks]

Find all functions g such that $g'(x) = 2 \sin(5x) + 2e^{5x} - \frac{2}{x} + \frac{3x^5 - \sqrt{x}}{x} + 2$.

ANSWER

$$g(x) = -\frac{2}{5} \cos(5x) + \frac{2}{5} e^{5x} - 2 \ln|x| + \frac{3}{5} x^5 - 2\sqrt{x} + 2x + C$$

RULES YOU NEED

- $\int \sin(ax) \, dx = -(1/a) \cos(ax)$, and $\int e^{ax} \, dx = (1/a) e^{ax}$.
- $\int x^n \, dx = x^{n+1}/(n+1)$ ($n \neq -1$), and $\int (1/x) \, dx = \ln|x|$.
- $\sqrt{x} = x^{1/2}$, and dividing powers means subtracting the powers.
- "All functions" means add $+ C$.

WORKING — STEP BY STEP

1 Split the fraction

Divide each term on top by x . For the root, $x^{1/2} \div x^1 = x^{1/2-1} = x^{-1/2}$:

$$\frac{3x^5 - \sqrt{x}}{x} = \frac{3x^5}{x} - \frac{x^{1/2}}{x} = 3x^4 - x^{-1/2}$$

$$g'(x) = 2 \sin(5x) + 2e^{5x} - \frac{2}{x} + 3x^4 - x^{-1/2} + 2$$

2 Integrate the sin and exponential terms

Here $a = 5$ in both, so divide by 5:

$$2 \sin(5x) \rightarrow 2 \times \left(-\frac{1}{5} \cos(5x)\right) = -\frac{2}{5} \cos(5x)$$

$$2e^{5x} \rightarrow 2 \times \frac{1}{5} e^{5x} = \frac{2}{5} e^{5x}$$

3 Integrate $2/x$

$$-2 \cdot (1/x) \rightarrow -2 \ln|x|$$

4 Integrate the powers

Add 1 to the power, divide by the new power. ($-1/2 + 1 = 1/2$, and $\div 1/2$ means $\times 2$):

$$3x^4 \rightarrow \frac{3x^5}{5} \quad -x^{-1/2} \rightarrow -\frac{x^{1/2}}{1/2} = -2x^{1/2} = -2\sqrt{x} \quad 2 \rightarrow 2x$$

5 Put it together, + C

$$g(x) = -\frac{2}{5}\cos(5x) + \frac{2}{5}e^{5x} - 2\ln|x| + \frac{3}{5}x^5 - 2\sqrt{x} + 2x + C$$

EXAM TECHNIQUE

Split $(3x^5 - \sqrt{x})/x$ into separate terms first. There is no rule for integrating a fraction like this as it stands.

Marks	Awarded for
2	$(3x^5 - \sqrt{x})/x$ split into $3x^4 - x^{-1/2}$.
1.5	$2\sin(5x) \rightarrow -(2/5)\cos(5x)$.
1.5	$2e^{5x} \rightarrow (2/5)e^{5x}$.
1	$-2/x \rightarrow -2\ln x $.
2.5	$(3/5)x^5 - 2\sqrt{x} + 2x$.
1	+ C included.

1(b)**[8 marks]**Find the function f given that $f''(x) = -3 + 10x - 12x^2$, $f(0) = 5$ and $f'(0) = 12$.**ANSWER**

$$f(x) = -x^4 + \frac{5}{3}x^3 - \frac{3}{2}x^2 + 12x + 5$$

RULES YOU NEED

- $\int x^n dx = x^{n+1}/(n+1)$.
- Integrate f'' to get f' (constant C), then f' to get f (constant D).
- $f'(0) = 12$ finds C; $f(0) = 5$ finds D.

WORKING — STEP BY STEP**1 Integrate $f''(x)$ to get $f'(x)$**

$$-3 \rightarrow -3x \quad 10x \rightarrow 10 \cdot \frac{x^2}{2} = 5x^2 \quad -12x^2 \rightarrow -12 \cdot \frac{x^3}{3} = -4x^3$$

$$f'(x) = -3x + 5x^2 - 4x^3 + C$$

2 Use $f'(0) = 12$ Every x term is 0 at $x = 0$:

$$f'(0) = C = 12$$

3 Integrate $f'(x)$ to get $f(x)$

$$-3x \rightarrow -\frac{3}{2}x^2 \quad 5x^2 \rightarrow \frac{5}{3}x^3 \quad -4x^3 \rightarrow -x^4 \quad 12 \rightarrow 12x$$

$$f(x) = -\frac{3}{2}x^2 + \frac{5}{3}x^3 - x^4 + 12x + D$$

4 Use $f(0) = 5$

$$f(0) = D = 5$$

5 Write it out (highest power first)

$$f(x) = -x^4 + \frac{5}{3}x^3 - \frac{3}{2}x^2 + 12x + 5$$

EXAM TECHNIQUEFind C before integrating the second time. Otherwise the 12 turns into $12x$ and is easy to lose.

Marks	Awarded for
2	$f'(x) = -3x + 5x^2 - 4x^3 + C.$
1.5	$C = 12.$
2.5	Integrating again correctly.
1.5	$D = 5.$
0.5	Final $f(x)$ stated.
Question 1 Total: 17.5 Marks	

Question 2

Partial fractions, integration by substitution — 17.5 marks

2(a)

[9.5 marks]

Find $\int \frac{3x + 11}{(x + 2)(x - 3)} dx$ by using the method of partial fractions.

ANSWER

$$-\ln|x + 2| + 4 \ln|x - 3| + C$$

RULES YOU NEED

- Write the fraction as $A/(x + 2) + B/(x - 3)$.
- Find A and B by putting in the x-values that make each bracket zero.
- $\int 1/(x - a) dx = \ln|x - a|$.

WORKING — STEP BY STEP

1 Set up the partial fractions

Multiply both sides by $(x + 2)(x - 3)$:

$$\frac{3x + 11}{(x + 2)(x - 3)} = \frac{A}{x + 2} + \frac{B}{x - 3} \Rightarrow 3x + 11 = A(x - 3) + B(x + 2)$$

2 Find B: put $x = 3$

The A bracket becomes 0:

$$3(3) + 11 = B(5) \Rightarrow 20 = 5B \Rightarrow B = 4$$

3 Find A: put $x = -2$

The B bracket becomes 0:

$$3(-2) + 11 = A(-5) \Rightarrow 5 = -5A \Rightarrow A = -1$$

Quick check: $A + B = 3$, the coefficient of x on top ✓.

4 Integrate each fraction

$$\int \left(\frac{-1}{x + 2} + \frac{4}{x - 3} \right) dx = -\ln|x + 2| + 4 \ln|x - 3| + C$$

EXAM TECHNIQUE

Choose x-values that make a bracket zero ($x = 3$ and $x = -2$). Each one kills a term and gives A or B straight away.

Marks	Awarded for
2	$3x + 11 = A(x - 3) + B(x + 2)$ set up.
2	$A = -1$.
2	$B = 4$.
2.5	Integrated to $-\ln x + 2 + 4 \ln x - 3 $.
1	$+ C$.

2(b)**[8 marks]**

Use Integration by Substitution to evaluate $\int 5x\sqrt{1-x^2} \, dx$.

ANSWER

$$-\frac{5}{3}(1-x^2)^{3/2} + C$$

RULES YOU NEED

- Substitution: let u = the inside of the root; $du = (du/dx) \, dx$.
- $\int u^{1/2} \, du = u^{3/2}/(3/2) = (2/3)u^{3/2}$.

WORKING — STEP BY STEP**1 Choose u**

Take the inside of the root. Its derivative, $-2x$, matches the x outside:

$$u = 1 - x^2$$

2 Find du and make $x \, dx$ on its own

$$\frac{du}{dx} = -2x \Rightarrow du = -2x \, dx \Rightarrow x \, dx = -\frac{1}{2} \, du$$

3 Substitute

The 5 stays in front:

$$\int 5\sqrt{u} \cdot \left(-\frac{1}{2}\right) \, du = -\frac{5}{2} \int u^{1/2} \, du$$

4 Integrate

$$-\frac{5}{2} \cdot \frac{2}{3} u^{3/2} + C = -\frac{5}{3} u^{3/2} + C$$

5 Put x back in

$$-\frac{5}{3}(1-x^2)^{3/2} + C$$

EXAM TECHNIQUE

Watch the minus sign: $du = -2x \, dx$, so $x \, dx = -(1/2) \, du$. Dropping it gives the answer with the wrong sign.

Marks	Awarded for
2.5	$u = 1 - x^2$, $du = -2x \, dx$.
1.5	Rewritten as $-(5/2) \int u^{1/2} \, du$.
2.5	Integrated: $-(5/3)u^{3/2}$.
1.5	Back in terms of x , $+ C$.
Question 2 Total: 17.5 Marks	

Question 3

Area between two curves, tangent slopes — 17.5 marks

3(a)

[9.5 marks]

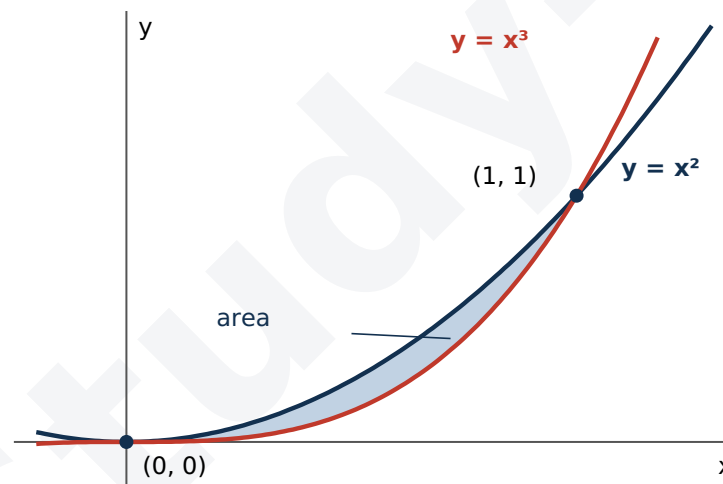
Find the area of the bounded region enclosed by the parabola $y = x^2$ and the graph of $y = x^3$.

ANSWER

Area = $\frac{1}{12}$ square units

RULES YOU NEED

- Area between two curves = $\int$ (upper curve – lower curve) dx .
- The limits are the x -values where the curves meet.



Between $x = 0$ and $x = 1$, $y = x^2$ is above $y = x^3$. The shaded sliver is the area.

WORKING — STEP BY STEP

1 Find where they meet

Set equal, bring to one side, factor out x^2 :

$$x^2 = x^3 \Rightarrow x^3 - x^2 = 0 \Rightarrow x^2(x - 1) = 0$$

$$x = 0 \quad \text{or} \quad x = 1$$

2 Which is on top?Test $x = \frac{1}{2}$:

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} \quad \left(\frac{1}{2}\right)^3 = \frac{1}{8} \Rightarrow y = x^2 \text{ on top}$$

3 Set up (upper – lower)

$$A = \int_0^1 (x^2 - x^3) dx$$

4 Integrate

$$A = \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1$$

5 Put in the limits

$$\left(\frac{1}{3} - \frac{1}{4}\right) - 0 = \frac{4}{12} - \frac{3}{12} = \frac{1}{12}$$

EXAM TECHNIQUE

Between 0 and 1, a higher power is **smaller** (e.g. $0.5^3 < 0.5^2$). Test a point rather than guessing which curve is on top.

Marks	Awarded for
2	Intersections $x = 0$ and $x = 1$.
2	$y = x^2$ identified as the upper curve.
2	$A = \int_0^1 (x^2 - x^3) dx$.
2	Integrated: $x^3/3 - x^4/4$.
1.5	$A = 1/12$.

3(b)**[8 marks]**

The curve for which $f(x) = e^{kx}$, where k is constant, is such that the tangent at $(1, e^2)$ passes through the origin. Find the slope of the tangent and hence determine the value of k .

ANSWER

Slope = e^2 , $k = 2$

NOTE

This part uses the slope of a tangent line (from MA1001) rather than an MA1002 integration topic. The only facts needed are: slope = rise ÷ run, and the slope of the tangent at $x = a$ is $f'(a)$.

RULES YOU NEED

- Slope through two points: $m = (y_2 - y_1)/(x_2 - x_1)$.
- The slope of the tangent to the curve at $x = a$ is $f'(a)$.
- If $e^k = e^2$, then $k = 2$.

WORKING — STEP BY STEP**1 Slope from the two points**

The tangent goes through the origin $(0, 0)$ and the point $(1, e^2)$:

$$\text{slope} = \frac{e^2 - 0}{1 - 0} = e^2$$

2 Slope from the derivative

The tangent at $x = 1$ has slope $f'(1)$:

$$f'(1) = e^{k(1)} = e^k$$

3 Set them equal and solve

Both are the slope of the same line:

$$e^k = e^2 \Rightarrow k = 2$$

(Taking $\ln$ of both sides: $k = \ln(e^2) = 2$.)

EXAM TECHNIQUE

You don't need $f(x)$ at all. Work out the slope from the two points, then match it to $f'(1)$.

Marks	Awarded for
3	Slope of the tangent = e^2 (from $(0, 0)$ and $(1, e^2)$).
2	Slope = $f'(1) = e^k$.
3	$e^k = e^2$, so $k = 2$.

Question 3 Total: 17.5 Marks

Question 4

Integration by parts, volumes of rotation — 17.5 marks

4(a)

[9.5 marks]

Use Integration by Parts to find $\int t^2 \sin(8t) dt$.

ANSWER

$$-\frac{t^2}{8} \cos(8t) + \frac{t}{32} \sin(8t) + \frac{1}{256} \cos(8t) + C$$

RULES YOU NEED

- Integration by parts: $\int u v' dt = u v - \int u' v dt$.
- Let u = the power of t (it gets simpler each time you differentiate).
- $\int \sin(8t) dt = -(1/8) \cos(8t)$ and $\int \cos(8t) dt = (1/8) \sin(8t)$.

WORKING — STEP BY STEP

1 Choose u and v'

$$u = t^2 \Rightarrow u' = 2t \quad v' = \sin(8t) \Rightarrow v = -\frac{1}{8} \cos(8t)$$

2 Apply the parts formula (first time)

Minus times minus gives plus:

$$\int t^2 \sin(8t) dt = -\frac{t^2}{8} \cos(8t) - \int 2t \cdot \left(-\frac{1}{8} \cos(8t)\right) dt$$

$$= -\frac{t^2}{8} \cos(8t) + \frac{1}{4} \int t \cos(8t) dt$$

3 Parts again on $\int t \cos(8t) dt$

$$u = t \Rightarrow u' = 1 \quad v' = \cos(8t) \Rightarrow v = \frac{1}{8} \sin(8t)$$

$$\int t \cos(8t) dt = \frac{t}{8} \sin(8t) - \int \frac{1}{8} \sin(8t) dt = \frac{t}{8} \sin(8t) + \frac{1}{64} \cos(8t)$$

($\int (1/8) \sin(8t) dt = (1/8)(-1/8) \cos(8t) = -(1/64) \cos(8t)$, and minus that is +.)

4 Put it back into step 2

Multiply by $1/4$: $(1/4)(t/8) = t/32$ and $(1/4)(1/64) = 1/256$:

$$-\frac{t^2}{8} \cos(8t) + \frac{t}{32} \sin(8t) + \frac{1}{256} \cos(8t) + C$$

EXAM TECHNIQUE

Keep track of the fractions: $1/8$ appears every time you integrate $\sin(8t)$ or $\cos(8t)$. Writing each step in brackets stops sign slips.

Marks	Awarded for
2	$u = t^2, v = -(1/8) \cos(8t)$.
2	First application: $-(t^2/8) \cos(8t) + (1/4) \int t \cos(8t) dt$.
2.5	Second application with $u = t, v = (1/8) \sin(8t)$.
1.5	$\int \sin(8t) dt$ handled correctly.
1.5	Final answer combined correctly, $+ C$.

4(b)**[8 marks]**

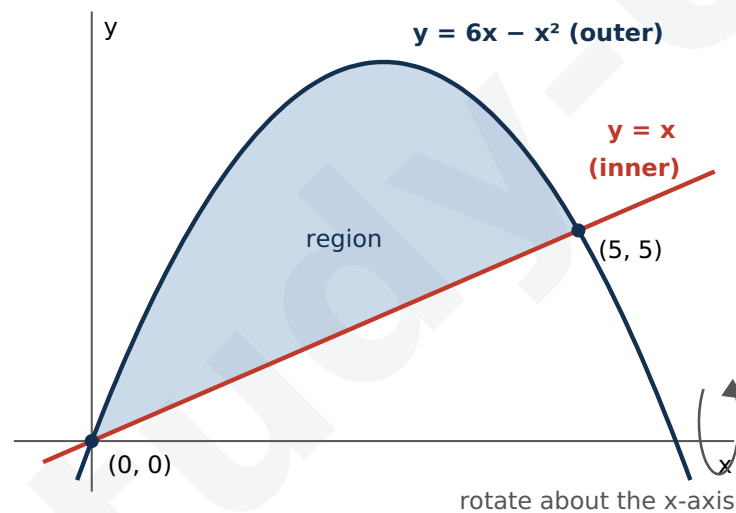
Calculate the volume generated by rotating around the x-axis the region bounded by the graphs of $y = 6x - x^2$ and $y = x$.

ANSWER

$$V = \frac{625\pi}{3} \text{ cubic units } (\approx 654.5)$$

RULES YOU NEED

- Volume about the x-axis: $V = \int_a^b \pi y^2 dx$.
- For a region between two curves: $V = \pi \int (y_{\text{outer}}^2 - y_{\text{inner}}^2) dx$.
- $(a - b)^2 = a^2 - 2ab + b^2$.



The region between the parabola and the line spins around the x-axis, making a solid with a hollow core.

WORKING — STEP BY STEP**1 Find where they meet**

$$6x - x^2 = x \Rightarrow 5x - x^2 = 0 \Rightarrow x(5 - x) = 0 \Rightarrow x = 0 \text{ or } x = 5$$

2 Which is further from the axis?

Test $x = 1$:

$$6(1) - 1^2 = 5 \quad y = 1 \Rightarrow \text{outer: } y = 6x - x^2, \text{ inner: } y = x$$

3 Set up the volume

Use $V = \int \pi y^2 dx$ for each curve and subtract the hole:

$$V = \pi \int_0^5 [(6x - x^2)^2 - x^2] dx$$

4 Expand the square

$$(6x - x^2)^2 = 36x^2 - 12x^3 + x^4$$

$$(6x - x^2)^2 - x^2 = 35x^2 - 12x^3 + x^4$$

5 Integrate

$$V = \pi \left[\frac{35x^3}{3} - 3x^4 + \frac{x^5}{5} \right]_0^5$$

6 Put in the limits

At $x = 5$: $5^3 = 125$, $5^4 = 625$, $5^5 = 3125$:

$$\frac{35 \times 125}{3} - 3 \times 625 + \frac{3125}{5} = \frac{4375}{3} - 1875 + 625 = \frac{4375}{3} - 1250$$

$$= \frac{4375}{3} - \frac{3750}{3} = \frac{625}{3} \Rightarrow V = \frac{625\pi}{3} \approx 654.5$$

EXAM TECHNIQUE

Square each curve separately: $(6x - x^2)^2 - x^2$, not $(6x - x^2 - x)^2$. The second one gives the wrong volume.

Marks	Awarded for
1.5	Intersections $x = 0$ and $x = 5$.
1	Outer $y = 6x - x^2$, inner $y = x$.
2	$V = \pi \int_0^5 [(6x - x^2)^2 - x^2] dx$ set up.
1	Expanded to $35x^2 - 12x^3 + x^4$.
1.5	Integrated: $35x^3/3 - 3x^4 + x^5/5$.
1	$V = 625\pi/3$.

Question 4 Total: 17.5 Marks

Question 5

Separable differential equations, least squares regression line — 17.5 marks

5(a)

[9.5 marks]

Use separation of variables to solve the differential equation $\frac{dy}{dx} = x^2y^2 + y^2 \cos x$ subject to the initial condition $y(0) = \frac{\pi}{4}$.

ANSWER

$$y = \frac{3\pi}{12 - \pi x^3 - 3\pi \sin x} \quad (\text{same as } y = \frac{1}{4/\pi - x^3/3 - \sin x})$$

RULES YOU NEED

- Separation of variables: all y terms with dy on one side, all x terms with dx on the other, then integrate.
- $\int y^{-2} dy = y^{-1}/(-1) = -1/y$.
- $\int \cos x dx = \sin x$, and $\sin 0 = 0$.

WORKING — STEP BY STEP

1 Factor out y^2

Both terms on the right have y^2 :

$$\frac{dy}{dx} = y^2(x^2 + \cos x)$$

2 Separate the variables

Divide by y^2 , multiply by dx :

$$\frac{1}{y^2} dy = (x^2 + \cos x) dx$$

3 Integrate both sides

$$\int y^{-2} dy = \int (x^2 + \cos x) dx \Rightarrow -\frac{1}{y} = \frac{x^3}{3} + \sin x + C$$

4 Use $y(0) = \pi/4$

Put $x = 0$, $y = \pi/4$. Note $1 \div (\pi/4) = 4/\pi$:

$$-\frac{4}{\pi} = 0 + \sin 0 + C = C \Rightarrow C = -\frac{4}{\pi}$$

5 Solve for y

Multiply both sides by -1 , then flip both sides:

$$\frac{1}{y} = \frac{4}{\pi} - \frac{x^3}{3} - \sin x \Rightarrow y = \frac{1}{4/\pi - x^3/3 - \sin x}$$

Multiply top and bottom by 3π : $y = \frac{3\pi}{12 - \pi x^3 - 3\pi \sin x}$

6 Check

$$y(0) = \frac{3\pi}{12 - 0 - 0} = \frac{\pi}{4} \checkmark$$

EXAM TECHNIQUE

Find C straight after integrating, while you still have $-1/y$. It's much easier than rearranging for y first.

Marks	Awarded for
2.5	Separated: $dy/y^2 = (x^2 + \cos x) dx$.
3	Integrated: $-1/y = x^3/3 + \sin x + C$.
2	$C = -4/\pi$ from $y(0) = \pi/4$.
2	$y = 3\pi/(12 - \pi x^3 - 3\pi \sin x)$ or equivalent.

5(b)**[8 marks]**

Find the least squares regression line $y = mx + b$, given the data points x : 0, 1, 2, 3 and y : 1, 0, -1, -1. Recall that the slope and intercept are given by $m = \frac{N\Sigma(xy) - \Sigma x \Sigma y}{N\Sigma(x^2) - (\Sigma x)^2}$ and $b = \frac{\Sigma y - m\Sigma x}{N}$. Give your answers in *fractional form*.

ANSWER

$$m = -\frac{7}{10}, \quad b = \frac{4}{5} \Rightarrow y = -\frac{7}{10}x + \frac{4}{5}$$

NOTE

Least squares regression isn't one of the listed MA1002 course topics. The question gives you both formulas, so it only needs careful arithmetic.

RULES YOU NEED

- Make a table with rows for x , y , xy and x^2 , then add up each row. N = number of points.
- Σx^2 means square each x , then add. $(\Sigma x)^2$ means add the x 's, then square.
- The question asks for fractions, so keep everything exact (no decimals).

WORKING — STEP BY STEP**1 Build the table of sums**

Here $N = 4$. Watch the negative y -values:

	1	2	3	4	Sum (Σ)
x	0	1	2	3	6
y	1	0	-1	-1	-1
xy	0	0	-2	-3	-5
x²	0	1	4	9	14

2 Top of the m formula

Minus a negative is a plus:

$$N\Sigma xy - \Sigma x \Sigma y = 4(-5) - (6)(-1) = -20 + 6 = -14$$

3 Bottom of the m formula

$$N\Sigma x^2 - (\Sigma x)^2 = 4(14) - 6^2 = 56 - 36 = 20$$

4 Find m

$$m = \frac{-14}{20} = -\frac{7}{10}$$

5 Find b

$$m\Sigma x = (-7/10)(6) = -42/10 = -21/5:$$

$$b = \frac{-1 - (-21/5)}{4} = \frac{-5/5 + 21/5}{4} = \frac{16/5}{4} = \frac{16}{20} = \frac{4}{5}$$

6 Write the line

$$y = -\frac{7}{10}x + \frac{4}{5}$$

EXAM TECHNIQUE

The signs catch people here: $\Sigma y = -1$ and $\Sigma xy = -5$. Write every negative in brackets when you substitute.

Marks	Awarded for
2	Sums $\Sigma x = 6$, $\Sigma y = -1$, $\Sigma xy = -5$, $\Sigma x^2 = 14$ (0.5 each).
3	$m = -14/20 = -7/10$.
2.5	$b = 4/5$.
0.5	Line $y = -(7/10)x + 4/5$ stated.

Question 5 Total: 17.5 Marks